

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

S.No	Register No.	Student Name	Assigned Question number
1	192311215	MUTHU VAISHNAVI T	1
2	192311229	JENIECATHRINA J	2
3	192324152	ABINAYA SRI N	3
4	192324287	PRAVEEN KUMAR R	4
5	192372122	NADENDLA THANVEER BASHA	5
6	192372336	SIGI SREENIVASULU	6
7	192411025	SRINIVASAN A	7
8	192411031	GUDIPALLI VEMULA SHABARISH	8
9	192411039	GOKUL RAJAN R	9
10	192411061	GANGADEVI LOKESH	10
11	192411156	KEERTHIPATI HEMA SREE	11
12	192411249	CHENNABOINA BHAVYA SRI	12
13	192421030	NAVEEN KUMAR S	13
14	192421126	MOHAMMED ANAS M	14
15	192421259	JAYAKRISHNA V	15
16	192421333	PUSHPAVALI V	16
17	192424016	DINESH M S	17
18	192424174	YOKESH KANNA R	18
19	192424259	PERA HEMIKA	19
20	192424327	CHARULATHA S	20
21	192424352	R SAI TEJA	21
22	192465048	POTHURAJU SAI CHARAN TEJ	22
23	192472087	MANCHIKALAPATI VISHNU VARDHAN	23
24	192472149	CHINNI VENKATA NAGA SAI POORNA TEJA	24
25	192472307	SHAIK AZEEM	25
26	192511060	Manoj M	26
27	192521075	Kaviya R	27
28	192521121	Srimathi S	28
29	192521352	Jagadish V	29
30	192524078	Rafeez Ahamed S I	30
31	192524237	Athin Siva S	31
32	192525202	Muthu Neelakanta Naidu	32
33	192525289	B Chandu	33
34	192525303	Yedururu Jithendra Reddy	34
35	192565016	Divya Sree P	35
36	192565069	Yaswanthika Shree S	36
37	192571005	V Sanjana	37
38	192571047	Shaffria A	38
39	192571051	Podalapalli Darshith	39
40	192572106	Theshon Mishmaan M	40

TECHNICAL PRESENTATION

Q1. Asymptotic Analysis of Sorting Algorithms

Analyze the efficiency of Merge Sort and Quick Sort using asymptotic notations. Clearly define the problem of comparing sorting algorithms for large datasets. Apply Big-O, Big-Theta, and Big-Omega notations to evaluate best, average, and worst cases. Use suitable tools (graphs or tables) to visualize performance differences. Present your findings clearly and justify which algorithm is more efficient under different scenarios.

Q2. Master Theorem Application

Consider the recurrence relation $T(n) = 2T(n/2) + n$. Analyze the problem using the Master Theorem and determine its time complexity. Clearly explain each case of the theorem and justify the selected case. Use tools such as recursion tree diagrams for better understanding. Present the solution clearly with proper interpretation of results.

Q3. Substitution Method Analysis

Analyze the recurrence $T(n) = T(n-1) + n$ using the substitution method. Clearly define the problem and apply step-by-step expansion. Justify each step mathematically and derive the final complexity. Use tables or diagrams if required. Present the analysis clearly and interpret the efficiency of the algorithm.

Q4. Comparison of Growth Rates

Compare different growth functions such as $\log n$, n , $n \log n$, n^2 , and 2^n . Analyze their impact on algorithm performance for large inputs. Use graphs or visualization tools to represent growth trends. Clearly explain which functions grow faster and justify your conclusions. Present insights on choosing efficient algorithms.

Q5. Recursion Tree Method

Analyze the recurrence $T(n) = 3T(n/2) + n$ using the recursion tree method. Clearly define the structure of the recursion tree. Compute the cost at each level and derive the total cost. Use diagrams for better explanation. Interpret the result and compare it with Master Theorem.

Q6. Best, Average, and Worst Case Analysis

Select an algorithm such as Linear Search or Binary Search and analyze its efficiency in best, average, and worst cases. Clearly define the problem and identify input conditions. Use asymptotic notations to represent each case. Present the results using tables or charts. Justify which case is most significant in real-world applications.

Q7. Space Complexity Analysis

Analyze both time and space complexity of Merge Sort. Clearly define the need for auxiliary space. Use asymptotic notation to evaluate memory usage. Compare it with an in-place algorithm like Quick Sort. Present findings using diagrams or tables. Interpret trade-offs between time and space efficiency.

Q8. Real-World Efficiency Analysis

A system processes large datasets using a specific algorithm. Analyze its efficiency using asymptotic notations. Clearly define the problem and input size considerations. Apply mathematical techniques to estimate performance. Use tools like graphs to visualize execution trends. Present conclusions on scalability and efficiency.

Q9. Divide and Conquer Efficiency

Analyze the efficiency of a Divide and Conquer algorithm using recurrence relations. Clearly define the problem and identify subproblems. Apply Master Theorem or substitution method to solve the recurrence. Use diagrams to explain the breakdown. Interpret the results and justify the efficiency of this approach.

Q10. Algorithm Comparison Using Visualization

Compare two algorithms of different complexities (e.g., $O(n^2)$ vs $O(n \log n)$) using graphical visualization. Clearly define the problem and dataset assumptions. Use tools like plotting graphs or tables to show differences. Analyze the results and justify why one algorithm outperforms the other. Present conclusions in a clear and professional manner.

Q11. Asymptotic Analysis of Searching Algorithms

Analyze the efficiency of Linear Search and Binary Search using asymptotic notations. Clearly define the problem of searching elements in datasets of varying sizes. Apply Big-O, Big-Theta, and Big-Omega notations to evaluate best, average, and worst cases. Use suitable tools such as tables or diagrams to illustrate how search space reduces. Present your findings clearly and justify which algorithm performs better for large datasets.

Q12. Master Theorem Application for Higher Recurrence

Consider the recurrence relation $T(n) = 3T(n/2) + n$. Analyze the problem using the Master Theorem and determine its time complexity. Clearly explain all cases of the theorem and justify the selected case. Use recursion tree diagrams or tabular representation for better understanding. Present the solution clearly with proper interpretation of results.

Q13. Substitution Method for Logarithmic Recurrence

Analyze the recurrence $T(n) = T(n/2) + \log n$ using the substitution method. Clearly define the recurrence and apply step-by-step expansion. Justify each step mathematically and derive the final complexity. Use tables or intermediate steps if required for clarity. Present the analysis clearly and interpret the efficiency of the algorithm.

Q14. Comparative Study of Growth Functions

Compare growth functions such as n , $n \log n$, n^2 , and 2^n for algorithm efficiency. Clearly define the problem of selecting efficient algorithms based on input size. Use graphical visualization or tabular comparison to represent growth rates. Explain which functions dominate others as n increases. Present insights on choosing optimal algorithms for large-scale problems.

Q15. Recursion Tree Method for Complexity Analysis

Analyze the recurrence $T(n) = 2T(n/2) + n \log n$ using the recursion tree method. Clearly construct the recursion tree and compute cost at each level. Sum the level costs to derive total complexity. Use diagrams or structured steps for clarity. Interpret the result and compare it with Master Theorem findings.

Q16. Case-Based Analysis of Algorithm Efficiency

Select an algorithm such as Merge Sort and analyze its efficiency under best, average, and worst-case scenarios. Clearly define input conditions and behavior in each case. Apply asymptotic notations to represent complexity. Use charts or tables for comparison. Justify which case is most relevant in practical applications.

Q17. Space Complexity Evaluation of Recursive Algorithms

Analyze the space complexity of Merge Sort and compare it with Quick Sort. Clearly define auxiliary space and recursion stack usage. Apply asymptotic notation to evaluate memory consumption. Use diagrams to represent recursion depth. Present findings and interpret trade-offs between time and space efficiency.

Q18. Real-Time Algorithm Efficiency Evaluation

A system processes large-scale input using an algorithm with complexity $O(n \log n)$. Analyze its efficiency using asymptotic notations. Clearly define input size assumptions and system constraints. Use graphical tools to visualize performance trends. Present conclusions on scalability and efficiency for real-world applications.

Q19. Divide and Conquer Recurrence Analysis

Analyze a divide-and-conquer algorithm with recurrence $T(n) = 4T(n/2) + n$. Clearly define subproblem structure and recurrence formulation. Apply Master Theorem to determine complexity. Use diagrams to explain recursive breakdown. Interpret the result and justify the efficiency of the approach.

Q20. Algorithm Comparison Using Graphical Representation

Compare two algorithms with complexities $O(n^2)$ and $O(n \log n)$ using graphical visualization. Clearly define the dataset and assumptions. Use graphs or tables to illustrate performance differences. Analyze results and explain why one algorithm outperforms the other. Present conclusions in a clear and professional manner.

Q21. Asymptotic Analysis of Recursive Searching Algorithms

Analyze the efficiency of recursive binary search using asymptotic notations. Clearly define the problem and explain how recursion reduces the input size at each step. Apply Big-O, Big-Theta, and Big-Omega to represent best, average, and worst cases. Use recursion diagrams or tables to illustrate the reduction process. Present the results clearly and justify the efficiency of the algorithm for large datasets.

Q22. Master Theorem for Non-Linear Recurrence

Consider the recurrence relation $T(n) = 5T(n/2) + n^2$. Analyze the problem using the Master Theorem and determine the time complexity. Clearly explain each case and justify the selected case based on comparison of functions. Use diagrams such as recursion trees for better understanding. Present the solution clearly and interpret the result.

Q23. Substitution Method for Polynomial Recurrence

Analyze the recurrence $T(n) = T(n-1) + n^2$ using the substitution method. Clearly define the problem and perform step-by-step expansion. Justify each transformation mathematically and derive the final complexity. Use tables or structured steps to illustrate growth. Present the analysis clearly and interpret the efficiency of the algorithm.

Q24. Comparative Growth Analysis of Algorithms

Compare algorithms with complexities $O(\log n)$, $O(n)$, $O(n \log n)$, and $O(n^2)$. Clearly define the problem of evaluating efficiency for large inputs. Use graphical visualization or tabular data to represent growth patterns. Explain which algorithm grows faster and justify your reasoning. Present conclusions on selecting efficient algorithms.

Q25. Recursion Tree Analysis for Complex Recurrence

Analyze the recurrence $T(n) = 3T(n/2) + n \log n$ using the recursion tree method. Clearly construct the tree and calculate the cost at each level. Sum the level costs to determine the total complexity. Use diagrams to explain the structure clearly. Interpret the result and compare it with Master Theorem.

Q26. Best, Average, and Worst Case in Sorting Algorithms

Select Quick Sort and analyze its efficiency under best, average, and worst-case scenarios. Clearly define input conditions and pivot selection strategy. Apply asymptotic notation for each case. Use charts or tables to compare performance. Justify which case significantly impacts real-world applications.

Q27. Space Complexity Analysis of Divide and Conquer Algorithms

Analyze the space complexity of Merge Sort focusing on recursion stack and auxiliary arrays. Clearly define memory requirements and usage. Apply asymptotic notation to represent space complexity. Compare with in-place algorithms such as Heap Sort. Present results using diagrams and interpret trade-offs.

Q28. Efficiency Analysis of Exponential Algorithms

Analyze an algorithm with complexity $O(2^n)$ and explain its performance limitations. Clearly define the problem and input size considerations. Use graphical tools to illustrate exponential growth. Compare with polynomial-time algorithms. Present conclusions on feasibility and scalability.

Q29. Divide and Conquer Algorithm Efficiency Evaluation

Analyze a divide-and-conquer algorithm with recurrence $T(n) = 2T(n/2) + n \log n$. Clearly define subproblem division and recurrence formulation. Apply Master Theorem to determine complexity. Use diagrams to show breakdown of recursive calls. Interpret the result and justify efficiency.

Q30. Algorithm Comparison Using Tabular Representation

Compare algorithms with complexities $O(n)$, $O(n \log n)$, and $O(n^2)$ using tabular representation. Clearly define dataset size and assumptions. Use tables to show performance differences. Analyze results and justify which algorithm performs best. Present conclusions in a clear manner.

Q31. Asymptotic Analysis of Iterative Algorithms

Analyze the efficiency of an iterative algorithm with nested loops. Clearly define the problem and count operations for each loop. Apply asymptotic notation to derive time complexity. Use structured steps or tables for clarity. Present the results and interpret the efficiency.

Q32. Master Theorem Application for Unequal Division

Consider the recurrence $T(n) = 2T(n/3) + n$. Analyze the problem using the Master Theorem. Clearly identify parameters and compare functions. Justify the selected case and derive the complexity. Use diagrams to support explanation and interpret the result.

Q33. Substitution Method for Logarithmic Growth

Analyze the recurrence $T(n) = T(n-1) + \log n$ using substitution method. Clearly define the recurrence and expand step-by-step. Justify each step mathematically and derive the final complexity. Use tables if needed to illustrate growth. Present results and interpret efficiency.

Q34. Growth Rate Comparison Using Graphs

Compare growth rates of $O(n)$, $O(n \log n)$, and $O(n^2)$ using graphical representation. Clearly define input range and assumptions. Plot graphs or use tables to visualize differences. Explain which grows faster and justify conclusions. Present insights for algorithm selection.

Q35. Recursion Tree Method for Multi-Branch Recurrence

Analyze the recurrence $T(n) = 4T(n/2) + n$ using recursion tree method. Clearly construct the tree and compute costs at each level. Sum the costs to find total complexity. Use diagrams for clarity. Interpret the result and compare with Master Theorem.

Q36. Case Analysis of Searching Algorithms

Analyze Binary Search under best, average, and worst-case scenarios. Clearly define conditions for each case. Apply asymptotic notation accordingly. Use diagrams or tables to illustrate behavior. Justify which case is most relevant in practice.

Q37. Space Complexity of Recursive vs Iterative Methods

Compare space complexity of recursive and iterative approaches for a problem. Clearly define stack usage and memory requirements. Apply asymptotic notation for both methods. Use diagrams to represent memory usage. Interpret trade-offs and justify preferred approach.

Q38. Efficiency Analysis of Polynomial Algorithms

Analyze an algorithm with complexity $O(n^3)$ and evaluate its scalability. Clearly define the problem and input size impact. Use graphical tools to show growth trend. Compare with lower-order complexities. Present conclusions on performance limitations.

Q39. Divide and Conquer Performance Comparison

Compare two divide-and-conquer algorithms with recurrences $T(n)=2T(n/2)+n$ and $T(n)=3T(n/2)+n$. Clearly define both recurrences. Apply Master Theorem to each. Use diagrams to compare recursive structures. Interpret which algorithm is more efficient and justify.

Q40. Algorithm Efficiency Using Visualization Tools

Analyze an algorithm's performance using graphical visualization. Clearly define complexity and input assumptions. Use graphs or tables to illustrate growth. Interpret the trend and identify efficiency. Present results clearly with justification.